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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

df= pd.read_csv(r"C:\Users\shrey\OneDrive - DIT University\Desktop\
ADST_Shreya\Data Set\claimants (1).csv")

df['CLMAGE'] = df['CLMAGE'].fillna(df['CLMAGE'].median())
df['CLMSEX'] = df['CLMSEX'].fillna(df['CLMSEX'].mode()[0])
df['CLMINSUR'] = df['CLMINSUR'].fillna(df['CLMINSUR'].mode()[0])
df['SEATBELT'] = df['SEATBELT'].fillna(df['SEATBELT'].mode()[0])

X = df[['CLMSEX', 'CLMINSUR', 'SEATBELT', 'CLMAGE', 'LOSS']]
y = df['ATTORNEY']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

LogisticRegression(max_iter=1000)
y_pred = model.predict(X_test)

print("Accuracy Score:")
print(accuracy_score(y_test, y_pred))

Accuracy Score:
0.7388059701492538

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

Confusion Matrix:
[[ 85  43]
 [ 27 113]]

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

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Classification Report:

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	precision	recall	f1-score	support
0	0.76	0.66	0.71	128

1	0.72	0.81	0.76	140
accuracy			0.74	268
macro avg	0.74	0.74	0.74	268
weighted avg	0.74	0.74	0.74	268

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from sklearn.model_selection import cross_val_score
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```
scores = cross_val_score(model, X, y, cv=5)
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```
print("\nCross Validation Scores:")
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```
print(scores)
```

```
print("\nMean Accuracy:")
```

```
print(scores.mean())
```

Cross Validation Scores:

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[0.68283582 0.69402985 0.69402985 0.71641791 0.7238806 ]
```

Mean Accuracy:

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0.7022388059701492
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